[INCH-POUND]

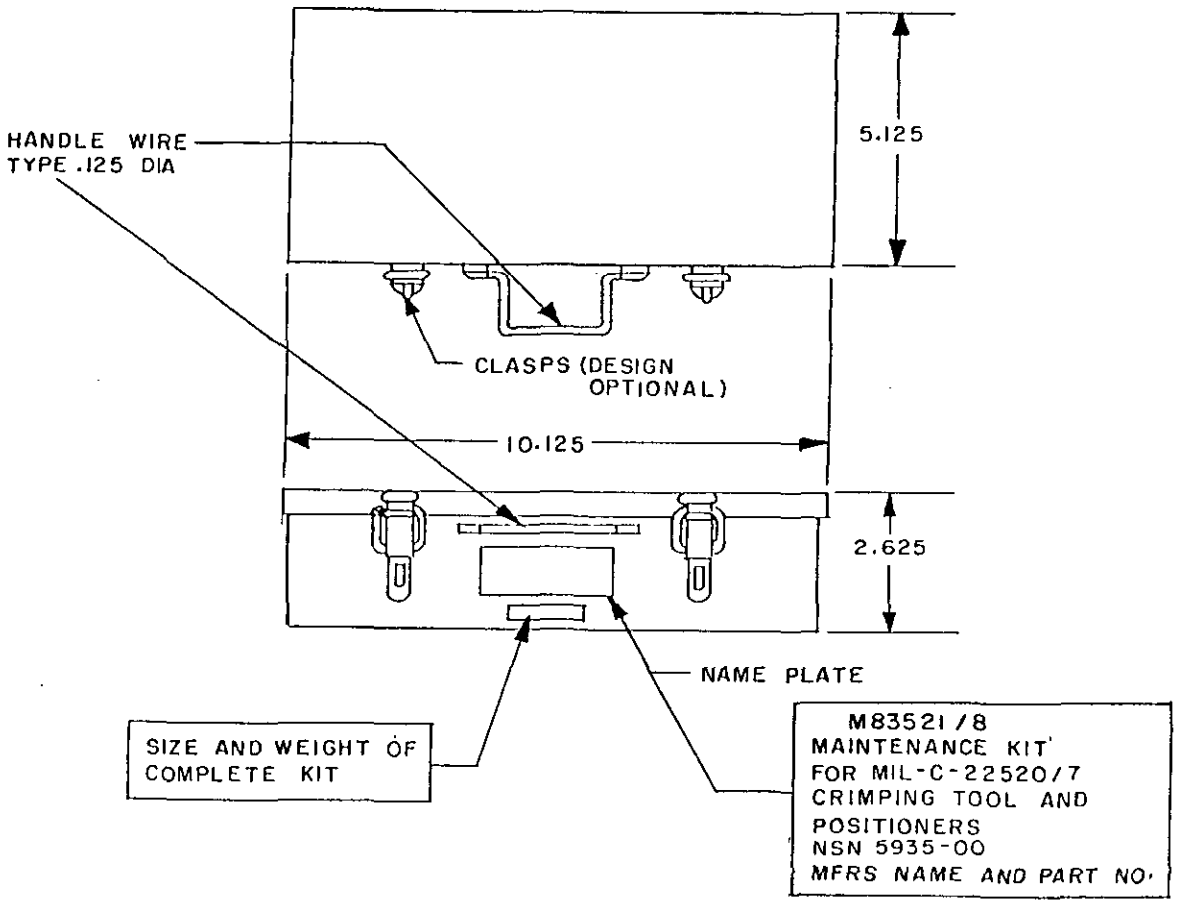
MIL-M-83521/8
25 April 1990
SUPERSEDING
MIL-T-83507/8
15 November 1977

MILITARY SPECIFICATION SHEET

MAINTENANCE KIT, ELECTRICAL CONTACT, CONNECTOR AND CABLE ACCESSORIES: MAINTENANCE KIT FOR MIL-C-22520/7 CRIMPING TOOL AND POSITIONERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-83521.



NOTES:
1. For drawing notes and metric equivalents, see figure 5.
2. Part number M83521/8-01 includes case assembly and tools.

FIGURE 1. Maintenance kit for MIL-C-22520/7 crimping tool and positioners.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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PARTS LIST:

(1) M22520/7-01 - Crimp tool frame.

(1) M22520/7-02 - Positioner.

(1) M22520/7-03 - Positioner.

(1) M22520/7-04 - Positioner.

(1) M22520/7-05 - Positioner.

(1) M22520/7-06 - Positioner.

(1) M22520/7-07 - Positioner.

(1) M22520/7-08 - Positioner.

(1) M22520/7-09 - Positioner.

(1) M22520/7-10 - Positioner.

(1) M22520/7-11 - Positioner.

(1) M22520/7-12 - Positioner.

(1) M22520/7-13 - Positioner.

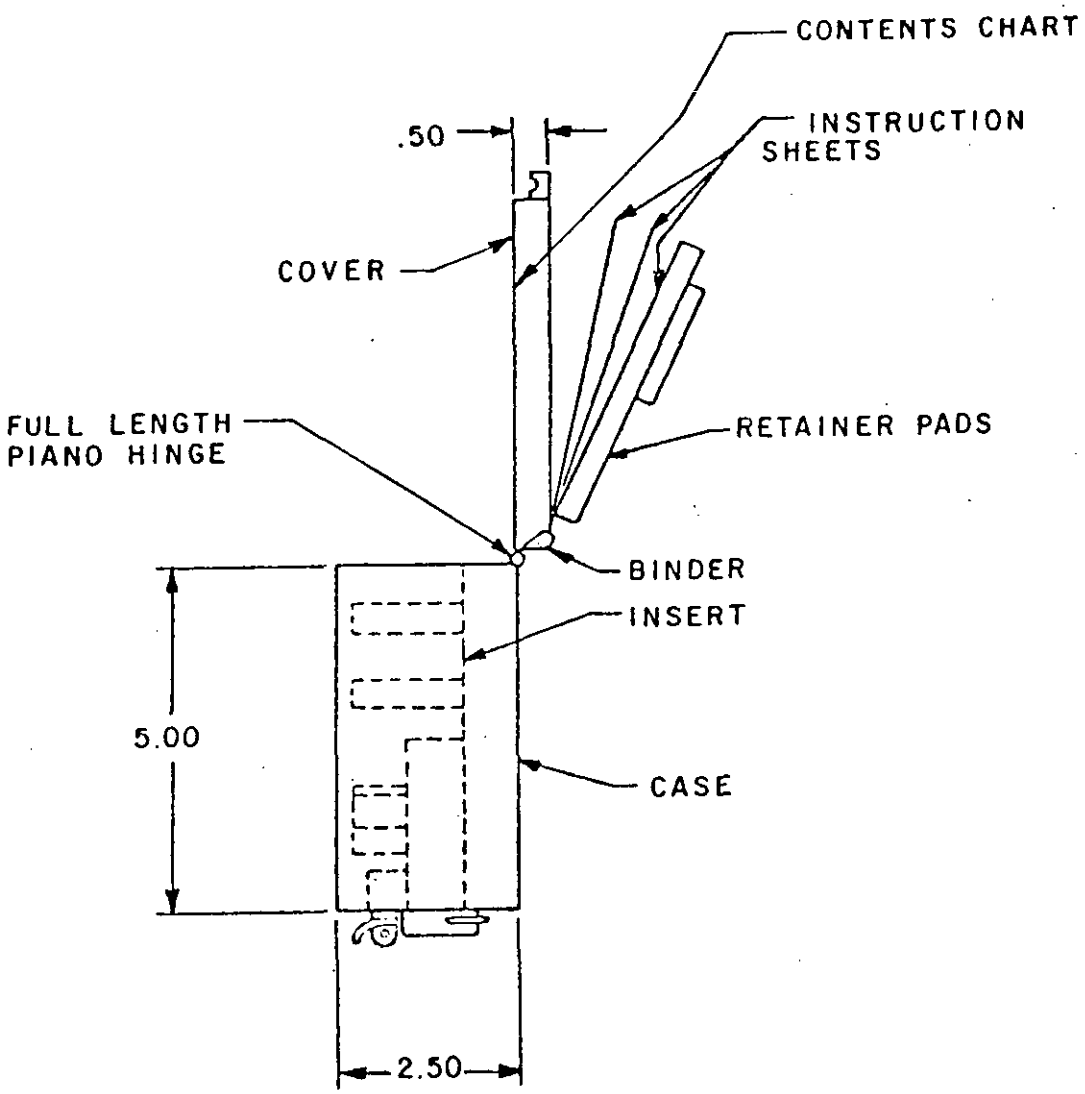
(1) M22520/3-3 - In-service inspection gauge.

(1) M83521/8-02 - Tool case assembly.

(1 set) Contents charts, instruction sheets, and tool selection chart.

FIGURE 1. Maintenance kit for MIL-C-22520/7 crimping tool and positioners - Continued.

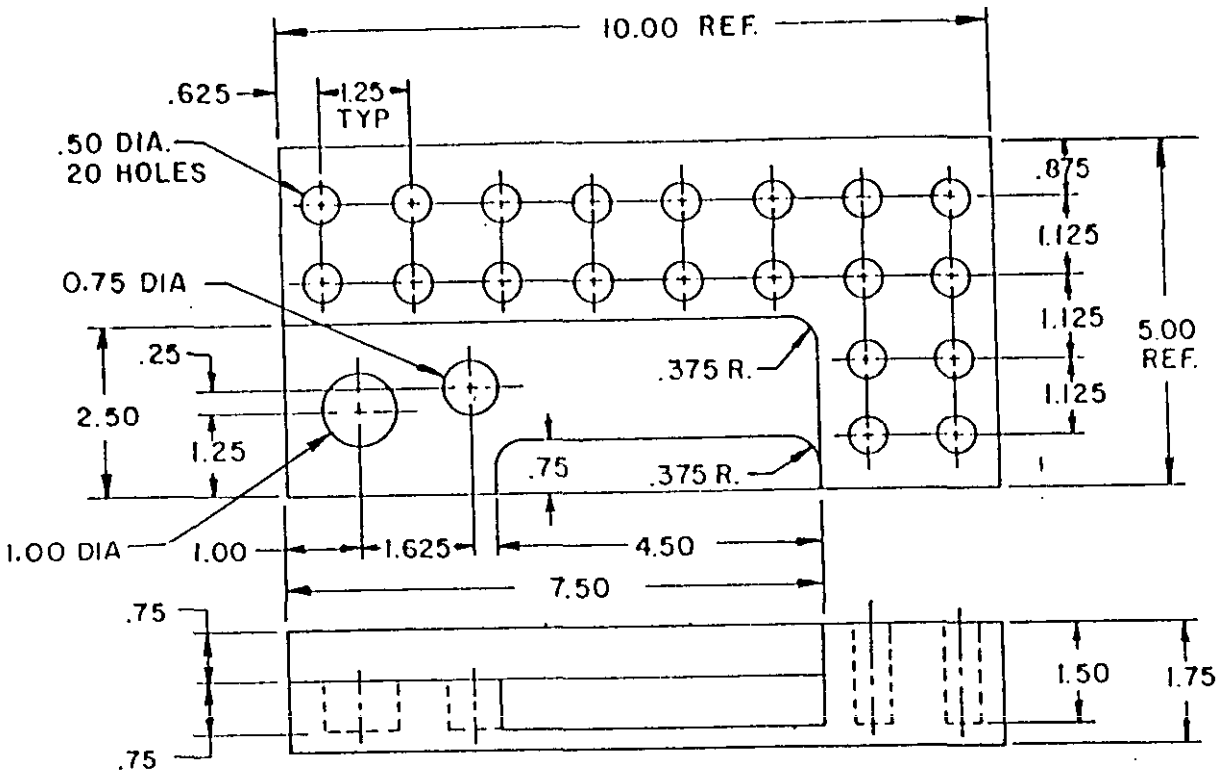
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NOTES:
1. See figure 5, note 4.
2. Part number M83521/8-02 case only (tools not included).

FIGURE 2. Tool case assembly.

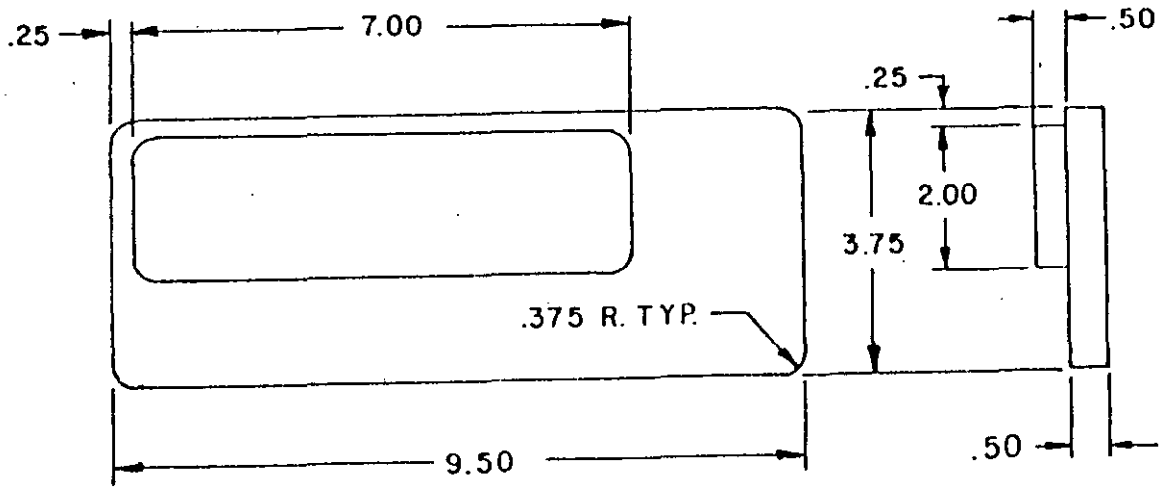
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NOTES:
1. For drawing notes and metric equivalents, see figure 5.
2. This insert is part of the tool case assembly (see figure 2).

FIGURE 3. Insert.

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NOTES:
1. For drawing notes and metric equivalents, see figure 5.
2. This retainer is part of the tool case assembly (see figure 2).

FIGURE 4. Retainer.

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Inches	mm
.125	3.18
.250	6.35
.375	9.52
.500	12.70
.625	15.88
.750	19.05
.875	22.22
1.000	25.40
1.125	28.58
1.250	31.75
1.500	38.10
1.625	41.28
1.750	44.45

Inches	mm
2.000	50.80
2.500	63.50
2.625	66.68
3.750	95.25
4.500	114.30
5.000	127.00
5.125	130.18
7.000	177.80
7.500	190.50
9.500	241.30
10.000	254.00
10.125	257.18

NOTES:
1. Dimensions are in iches.
2. Metric eqivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.031 (0.79 mm).
4. Remove all burrs, break sharp edges and corners.

FIGURE 5. Drawing notes and metric equivalents.

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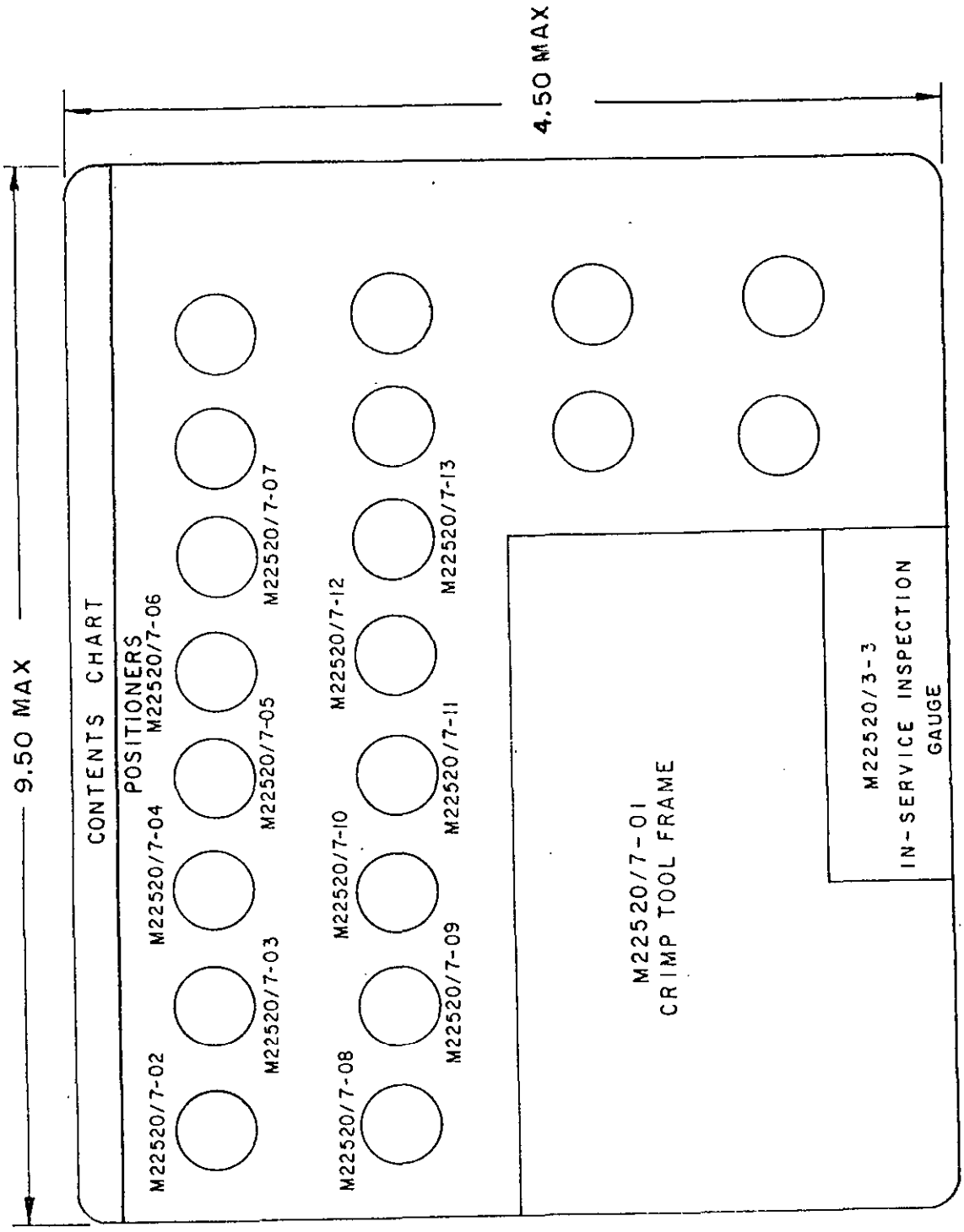


FIGURE 6. Contents chart.

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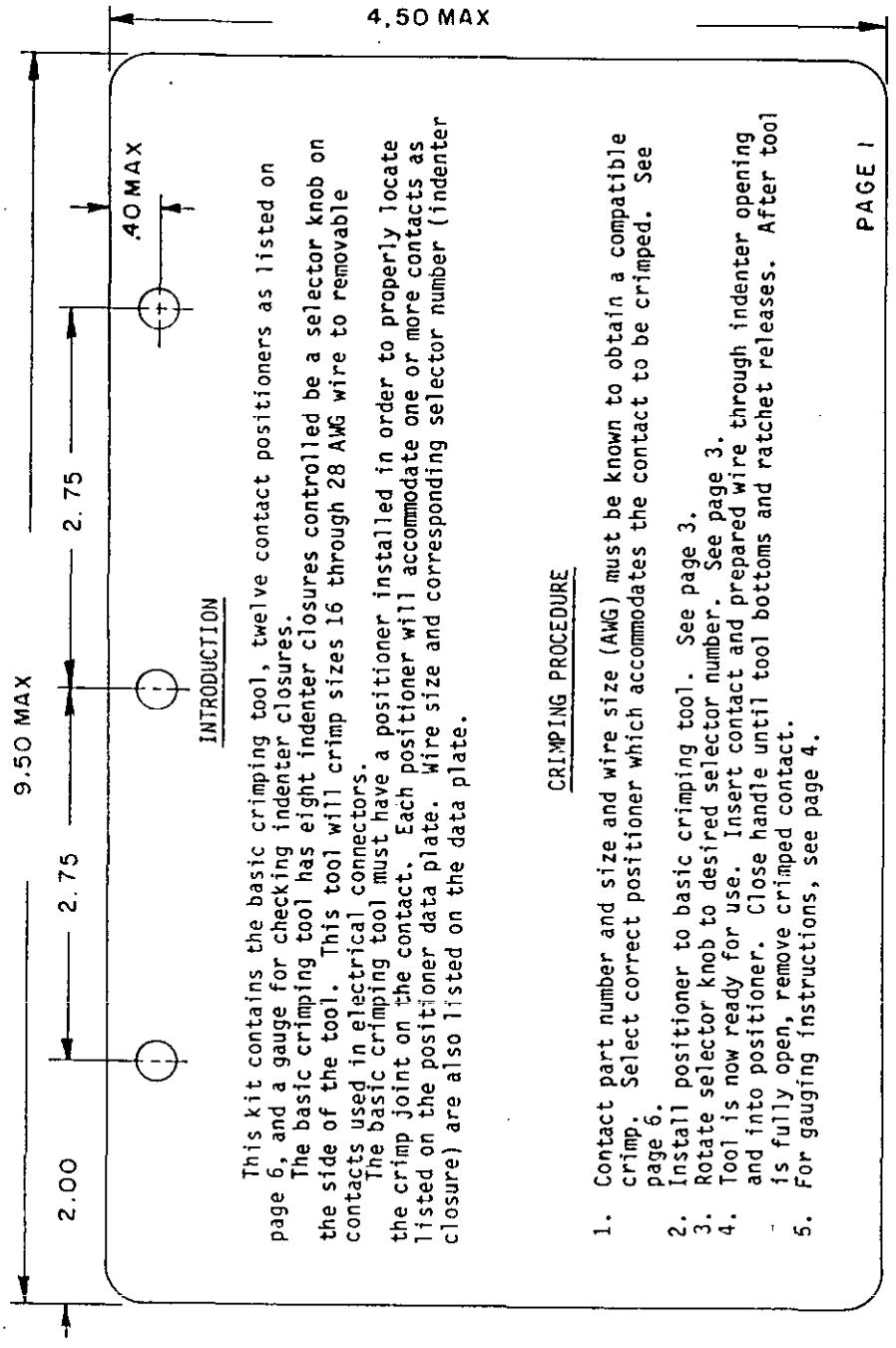


FIGURE 7. *Instruction sheet.*

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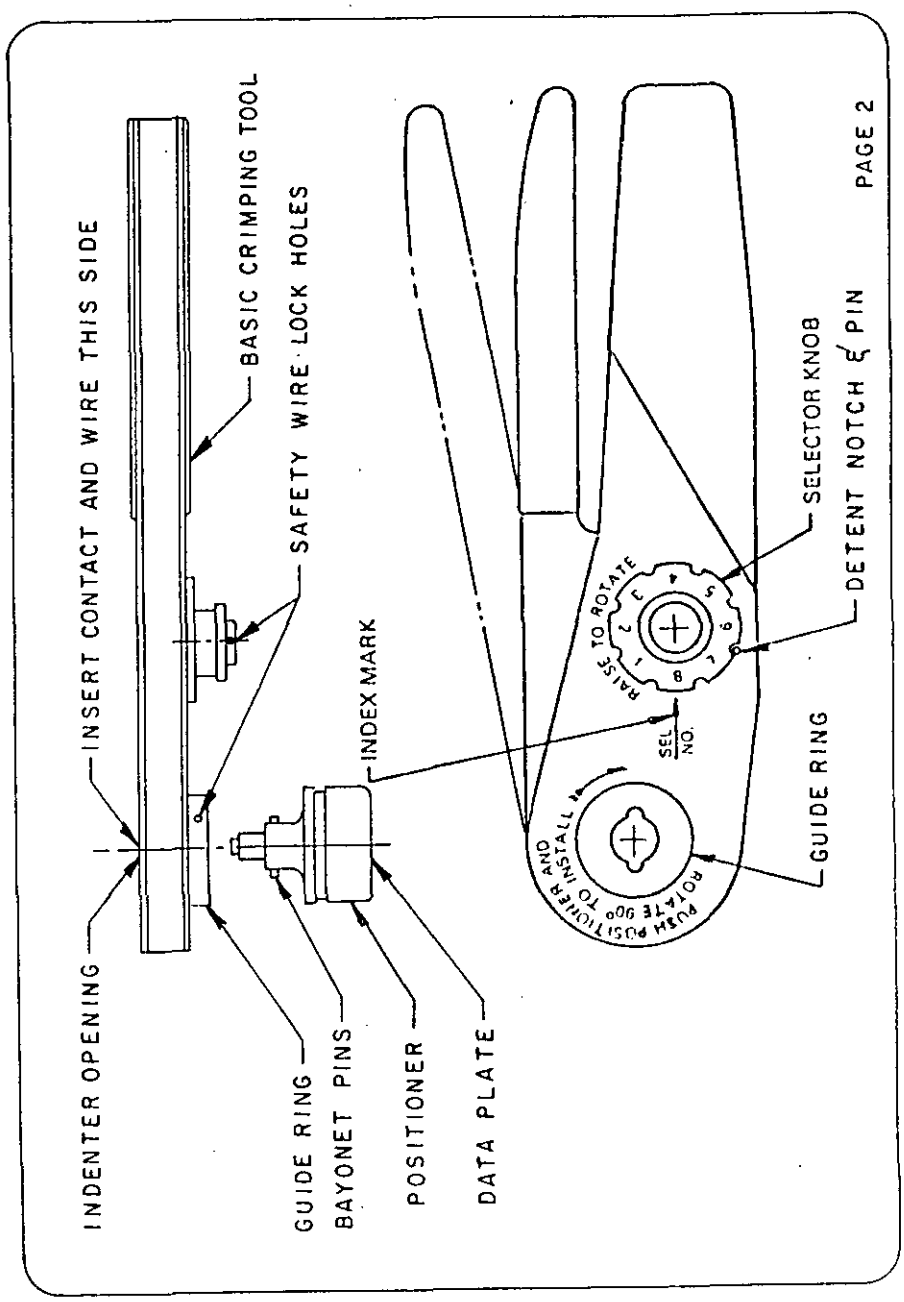
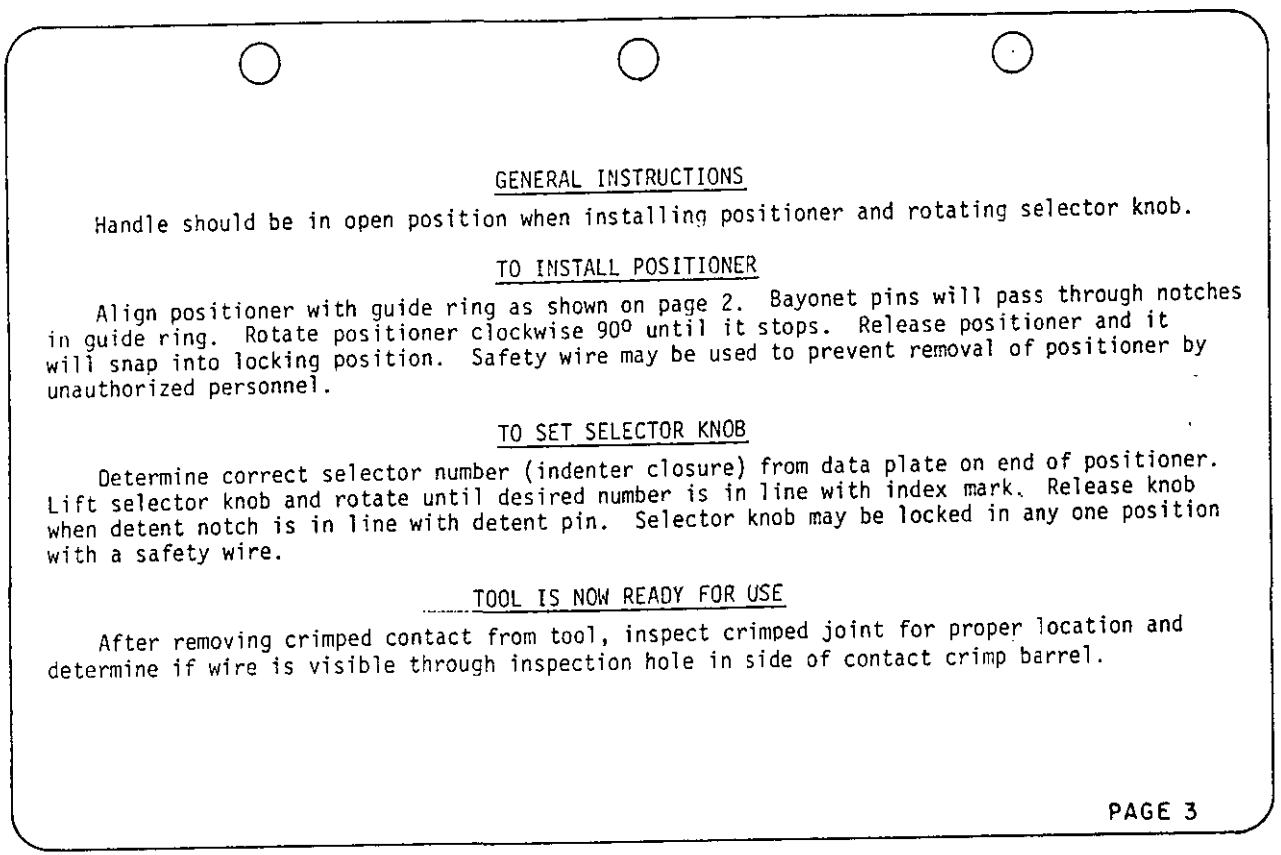


FIGURE 8. Tool.

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GENERAL INSTRUCTIONS

Handle should be in open position when installing positioner and rotating selector knob.

TO INSTALL POSITIONER

Align positioner with guide ring as shown on page 2. Bayonet pins will pass through notches in guide ring. Rotate positioner clockwise 90° until it stops. Release positioner and it will snap into locking position. Safety wire may be used to prevent removal of positioner by unauthorized personnel.

TO SET SELECTOR KNOB

Determine correct selector number (indenter closure) from data plate on end of positioner. Lift selector knob and rotate until desired number is in line with index mark. Release knob when detent notch is in line with detent pin. Selector knob may be locked in any one position with a safety wire.

TOOL IS NOW READY FOR USE

After removing crimped contact from tool, inspect crimped joint for proper location and determine if wire is visible through inspection hole in side of contact crimp barrel.

PAGE 3

FIGURE 9. General instructions.

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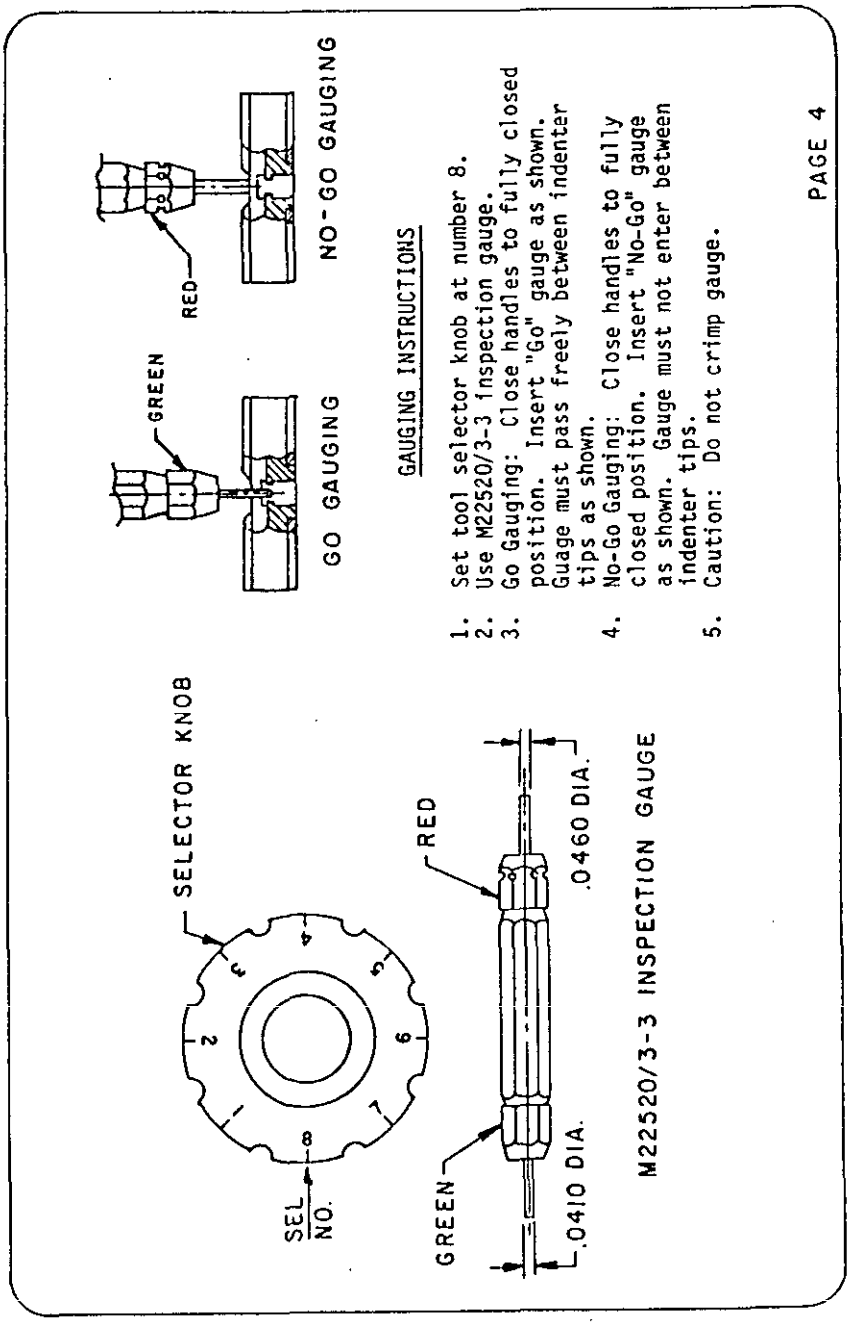
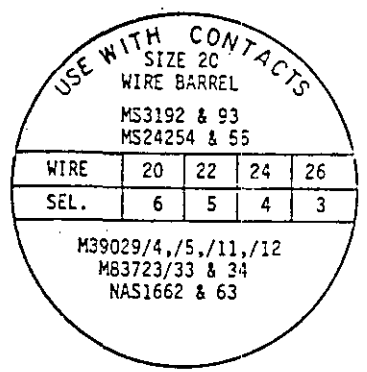
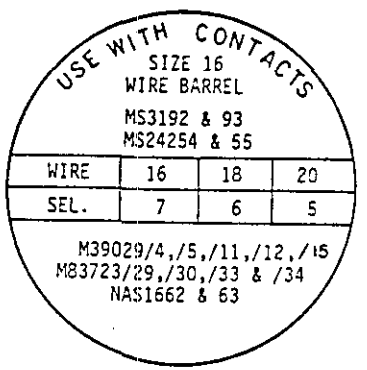


FIGURE 10. Gauging instructions sheet.

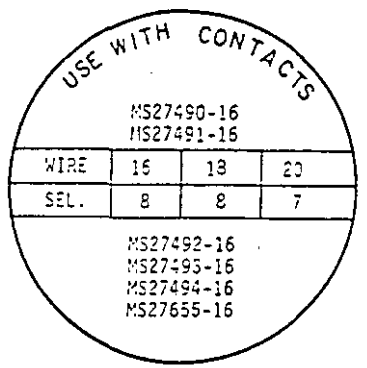
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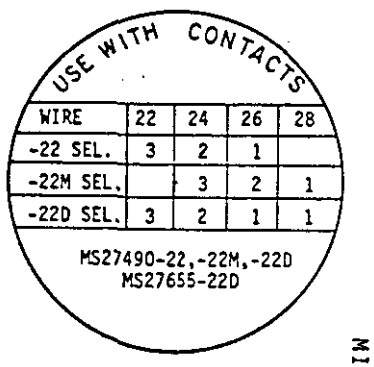
FOR M22520/7-02 POSITIONER



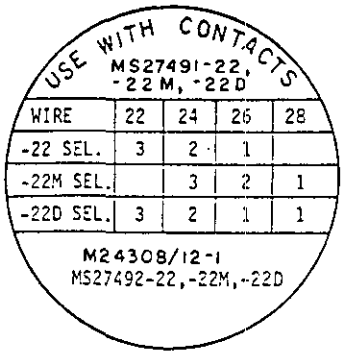
FOR M22520/7-03 POSITIONER



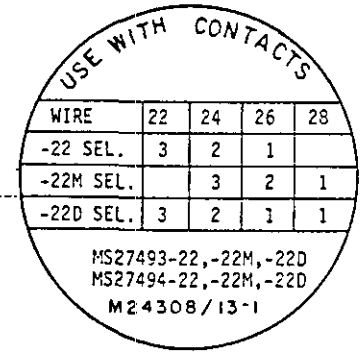
FOR M22520/7-04 POSITIONER



FOR M22520/7-05 POSITIONER



FOR M22520/7-06 POSITIONER



FOR M22520/7-07 POSITIONER

EACH DATA PLATE DEFINES THE CONTACT, AWG, AND SELECTOR NUMBER. CHECK POSITIONER FOR CORRECT DATA PLATE BEFORE USING.

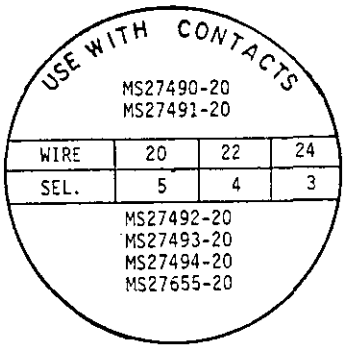
FIGURE 11. Positioners data plates.

POSITIONER DATA PLATES

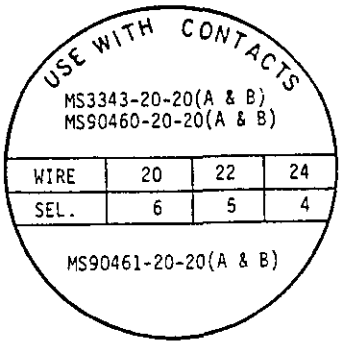
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POSITIONER DATA PLATES

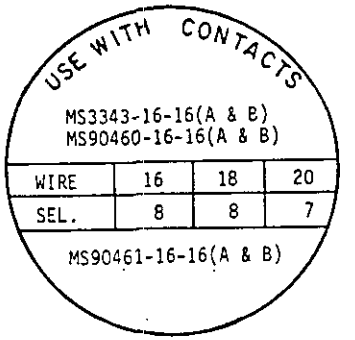
FIGURE 11. Positioners data plates - Continued.



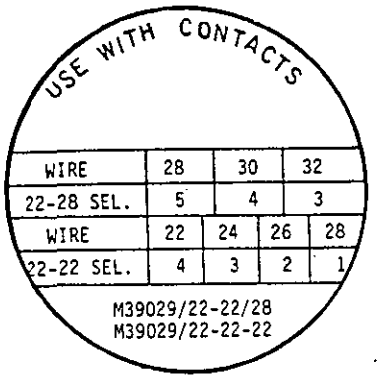
FOR M22520/7-08 POSITIONER



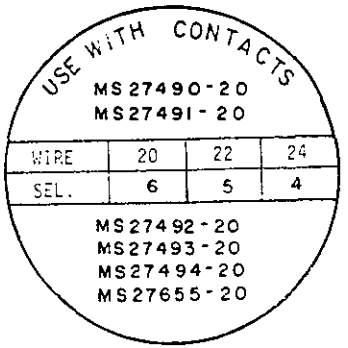
FOR M22520/7-09 POSITIONER



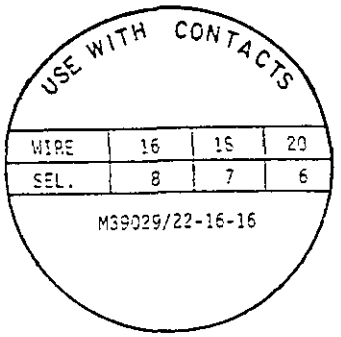
FOR M22520/7-10 POSITIONER



FOR M22520/7-11 POSITIONER



FOR M22520/7-12 POSITIONER



FOR M22520/7-13 POSITIONER

EACH DATA PLATE DEFINES THE CONTACT, AWG, AND SELECTOR NUMBER. CHECK POSITIONER FOR CORRECT DATA PLATE BEFORE USING.

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REQUIREMENTS:

Design and configuration: See figure 1 through 5.

Materials:

Case and cover (category 3): Sheet metal, 21 gauge in accordance with QQ-S-698; gray baked enamel on all metal parts.

Instruction sheets and contents chart: White vinyl .020 (0.51 mm) thick, or white paper sealed in weatherproof clear laminate .02 (0.051 mm).

Insert and retainer pad: Fire retardant formed polyethylene in accordance with PPP-C-1752; color is optional.

Clasps, draw pull type with provision for locking device, handles, and hinges: Suitably plated to resist corrosion.

Nameplate: In accordance with MIL-P-19834.

Binder: Looseleaf, three-ring, .562 (14.27 mm) ±.062 (1.57 mm), 2.75 (69.9 mm) center distance between rings, securely riveted to inside of cover.

Instruction sheets and contents chart (see page 6 through 11: Lettering shall be legible and protected with a transparent coating. Contents chart (centered on inside of case cover)) shall have pressure sensitive adhesive backing and show the insert cavity configurations and the respective tool part numbers.

Part number:

M83521/8-01 - Maintenance kit for MIL-C-22520/7 crimping tool and positioners.
M83521/8-02 - Tool case assembly.

Part number 1/	Superseded by
M83507/8-01	M83521/8-01
M83507/8-02	M83521/8-02

1/ Inactive for new design.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, CL, MI
Navy - EC
Air Force - 11, 99
DLA - ES

User activities:
Army - AT
Navy - MC, SH
Air Force - 19

Preparing activity:
Air Force - 85

Agent: DLA - ES

(Project 5935-3712)

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